

Exercise 2.2 Complex Machine Learning Models and Keras Part 1

CNN (Convolution Neural Network) Model

I began by using a CNN model because it runs quickly and handles large datasets efficiently. Although CNNs are typically used for spatial or image data, I wanted to explore how well it performs with time-series data like this, and plan to trial an RNN model as well.

Throughout testing, I experimented with various hyperparameters, including different epoch levels (10, 15, and 30), batch sizes (4, 16, and 32), hidden layer sizes (ranging from 2 to 256), and activation functions such as ReLU, softmax, sigmoid, and tanh. ReLU produced the highest accuracy at 64.4%, but the loss became NaN, suggesting overfitting. Tanh showed more stability with a lower but solid accuracy of 59.1% and a loss that stabilized at 26.79. Interestingly, softmax and sigmoid produced lower accuracy, but they were the only activations that allowed the model to correctly identify all 15 weather stations, with softmax performing best when epoch = 30, batch size = 16, and hidden layer size = 32. I also observed that increasing layer size and decreasing batch size slowed down training time. In some runs, the model didn't converge, with loss increasing rapidly and accuracy stuck around 12%. While softmax currently seems best for full station recognition, ReLU and tanh remain promising due to their higher accuracy and more stable loss values. Overall, the model setup is a solid starting point, and further tuning will help improve its performance for ClimateWins' needs.

Scenario 1:

Hyperparameters

Epoch = 30

Batch Size = 16

Hidden Layers = 32

Activation function: Softmax

```
epochs = 30
batch_size = 16
n_hidden = 32

timesteps = len(X_train[0])
input_dim = len(X_train[0][0])
n_classes = len(y_train[0])

model = Sequential()
model.add(Conv1D(n_hidden, kernel_size=2, activation='relu', input_shape=(timesteps, input_dim)))
model.add(Dense(16, activation='relu'))
model.add(MaxPooling1D())
model.add(Flatten())
model.add(Dense(n_classes, activation='softmax')) # Options: sigmoid, tanh, softmax, relu
```

Partial Output:

Epoch 1/30

1076/1076 – 1s – 792us/step – accuracy: 0.1189 – loss: 5041.5273
Epoch 2/30
1076/1076 – 0s – 457us/step – accuracy: 0.1437 – loss: 54822.8867
Epoch 3/30
1076/1076 – 1s – 478us/step – accuracy: 0.1351 – loss: 177908.4531
Epoch 4/30
1076/1076 – 0s – 456us/step – accuracy: 0.1335 – loss: 381350.8125
Epoch 5/30
1076/1076 – 1s – 465us/step – accuracy: 0.1349 – loss: 663748.0625
Epoch 6/30
1076/1076 – 0s – 457us/step – accuracy: 0.1358 – loss: 1087406.3750
Epoch 7/30
1076/1076 – 0s – 459us/step – accuracy: 0.1306 – loss: 1571165.1250
Epoch 8/30
1076/1076 – 0s – 452us/step – accuracy: 0.1310 – loss: 2174751.0000
Epoch 9/30
1076/1076 – 0s – 440us/step – accuracy: 0.1337 – loss: 2933341.7500
Epoch 10/30
1076/1076 – 0s – 453us/step – accuracy: 0.1307 – loss: 3841698.5000
Epoch 11/30
1076/1076 – 1s – 466us/step – accuracy: 0.1318 – loss: 4917219.5000
Epoch 12/30
1076/1076 – 0s – 446us/step – accuracy: 0.1299 – loss: 6044450.5000
Epoch 13/30
1076/1076 – 0s – 445us/step – accuracy: 0.1324 – loss: 7433755.0000
Epoch 14/30
1076/1076 – 0s – 446us/step – accuracy: 0.1318 – loss: 8950011.0000
Epoch 15/30
1076/1076 – 0s – 445us/step – accuracy: 0.1297 – loss: 10705555.0000

Confusion Matrix:

180/180		0s 407us/step					
Pred		BASEL	BELGRADE	BUDAPEST	DEBILT	DUSSELDORF	HEATHROW \
True							
BASEL		84	617	143	1089	49	32
BELGRADE		0	246	1	234	0	0
BUDAPEST		0	27	0	61	0	0
DEBILT		0	11	0	38	0	0
DUSSELDORF		0	2	0	17	0	0
HEATHROW		0	1	0	41	0	0
KASSEL		0	3	0	3	0	0
LJUBLJANA		0	12	0	12	0	0
MAASTRICHT		0	2	0	3	0	0
MADRID		1	21	3	154	0	1
MUNCHENB		0	2	1	0	0	0
OSLO		0	0	0	3	0	0
STOCKHOLM		0	2	0	1	0	0
VALENTIA		0	0	0	1	0	0

Pred		LJUBLJANA	MAASTRICHT	MADRID	MUNCHENB	OSLO	SONNBLICK \
True							
BASEL		32	10	1491	3	52	8
BELGRADE		0	0	604	0	7	0
BUDAPEST		0	0	123	0	3	0
DEBILT		0	0	32	0	1	0
DUSSELDORF		0	0	10	0	0	0
HEATHROW		0	0	37	0	3	0
KASSEL		0	0	4	0	1	0
LJUBLJANA		0	0	37	0	0	0
MAASTRICHT		0	0	4	0	0	0
MADRID		0	0	266	0	9	0
MUNCHENB		0	0	4	0	1	0
OSLO		0	0	1	0	1	0
STOCKHOLM		0	0	1	0	0	0
VALENTIA		0	0	0	0	0	0

Scenario 2:

Hyperparameters

Epochs = 30

Batch size = 16

Hidden Layers = 64

Activation function: tanh

```
epochs = 30
batch_size = 16
n_hidden = 64

timesteps = len(X_train[0])
input_dim = len(X_train[0][0])
n_classes = len(y_train[0])

model = Sequential()
model.add(Conv1D(n_hidden, kernel_size=2, activation='relu', input_shape=(timesteps, input_dim)))
model.add(Dense(16, activation='relu'))
model.add(MaxPooling1D())
model.add(Flatten())
model.add(Dense(n_classes, activation='tanh')) # Options: sigmoid, tanh, softmax, relu
```

Part of Output:

Epoch 1/30

1076/1076 - 1s - 957us/step - accuracy: 0.0188 - loss: 23.9314

Epoch 2/30

1076/1076 - 1s - 618us/step - accuracy: 0.0440 - loss: 24.6342

Epoch 3/30

1076/1076 - 1s - 623us/step - accuracy: 0.0494 - loss: 26.3881

Epoch 4/30

1076/1076 - 1s - 626us/step - accuracy: 0.0587 - loss: 26.8910

Epoch 5/30

1076/1076 - 1s - 620us/step - accuracy: 0.0627 - loss: 26.9181

Epoch 6/30

1076/1076 - 1s - 608us/step - accuracy: 0.0657 - loss: 26.9293

Epoch 7/30

1076/1076 - 1s - 623us/step - accuracy: 0.0686 - loss: 26.9415

Epoch 8/30

1076/1076 - 1s - 621us/step - accuracy: 0.0710 - loss: 26.9593

Epoch 9/30

1076/1076 - 1s - 620us/step - accuracy: 0.0746 - loss: 26.9649

Epoch 10/30

1076/1076 - 1s - 629us/step - accuracy: 0.0801 - loss: 26.9864

Epoch 11/30

1076/1076 - 1s - 618us/step - accuracy: 0.0833 - loss: 26.9865

Epoch 12/30

1076/1076 - 1s - 624us/step - accuracy: 0.0863 - loss: 26.9855

Epoch 13/30

1076/1076 - 1s - 627us/step - accuracy: 0.0914 - loss: 26.9968

Epoch 14/30

1076/1076 - 1s - 619us/step - accuracy: 0.0953 - loss: 27.0033

Epoch 15/30

1076/1076 - 1s - 626us/step - accuracy: 0.0966 - loss: 27.0033

Epoch 16/30

1076/1076 - 1s - 627us/step - accuracy: 0.0968 - loss: 27.0033

Epoch 17/30

1076/1076 - 1s - 624us/step - accuracy: 0.0968 - loss: 27.0033

Epoch 18/30

1076/1076 - 1s - 621us/step - accuracy: 0.0968 - loss: 27.0033

Epoch 19/30

Confusion Matrix:

180/180	0s 265us/step
Pred	BASEL
True	
BASEL	3682
BELGRADE	1092
BUDAPEST	214
DEBILT	82
DUSSELDORF	29
HEATHROW	82
KASSEL	11
LJUBLJANA	61
MAASTRICHT	9
MADRID	458
MUNCHENB	8
OSLO	5
STOCKHOLM	4
VALENTIA	1

Scenario 3:

Hyperparameters

Epochs = 15

Batch size = 4

Hidden Layers = 4

Activation function: ReLU

```
epochs = 15
batch_size = 4
n_hidden = 4

timesteps = len(X_train[0])
input_dim = len(X_train[0][0])
n_classes = len(y_train[0])

model = Sequential()
model.add(Conv1D(n_hidden, kernel_size=2, activation='relu', input_shape=(timesteps, input_dim)))
model.add(Dense(16, activation='relu'))
model.add(MaxPooling1D())
model.add(Flatten())
model.add(Dense(n_classes, activation='relu')) # Options: sigmoid, tanh, softmax, relu
```

Output:

Epoch 1/15

4303/4303 - 2s - 489us/step - accuracy: 0.0919 - loss: 18.4190

Epoch 2/15

4303/4303 - 2s - 384us/step - accuracy: 0.0906 - loss: 17.8615

Epoch 3/15

4303/4303 - 2s - 384us/step - accuracy: 0.0840 - loss: 17.3831

Epoch 4/15

4303/4303 - 2s - 393us/step - accuracy: 0.3812 - loss: nan

Epoch 5/15

4303/4303 - 2s - 380us/step - accuracy: 0.6440 - loss: nan

Epoch 6/15

4303/4303 - 2s - 374us/step - accuracy: 0.6440 - loss: nan

Epoch 7/15

4303/4303 - 2s - 376us/step - accuracy: 0.6440 - loss: nan

Epoch 8/15

4303/4303 - 2s - 374us/step - accuracy: 0.6440 - loss: nan

Epoch 9/15

4303/4303 - 2s - 374us/step - accuracy: 0.6440 - loss: nan
 Epoch 10/15
 4303/4303 - 2s - 375us/step - accuracy: 0.6440 - loss: nan
 Epoch 11/15
 4303/4303 - 2s - 375us/step - accuracy: 0.6440 - loss: nan
 Epoch 12/15
 4303/4303 - 2s - 375us/step - accuracy: 0.6440 - loss: nan
 Epoch 13/15
 4303/4303 - 2s - 375us/step - accuracy: 0.6440 - loss: nan
 Epoch 14/15
 4303/4303 - 2s - 376us/step - accuracy: 0.6440 - loss: nan
 Epoch 15/15
 4303/4303 - 2s - 375us/step - accuracy: 0.6440 - loss: nan

Confusion Matrix:

180/180	0s 351us/step
Pred	BASEL
True	
BASEL	3682
BELGRADE	1092
BUDAPEST	214
DEBILT	82
DUSSELDORF	29
HEATHROW	82
KASSEL	11
LJUBLJANA	61
MAASTRICHT	9
MADRID	458
MUNCHENB	8
OSLO	5
STOCKHOLM	4
VALENTIA	1

RNN (Recurrent Neural Network)/LSTM Model

The LSTM model takes more time to run compared to the CNN model. Using the sigmoid activation function with the current parameters results in very low performance, with only 5% accuracy, a loss of 20.20, and the ability to recognize just 2 out of 15 stations. When using tanh, the model runs even slower and performs slightly worse, achieving an accuracy of 4.37%, a loss of 23.95, and recognizing 6 stations. With Softmax activation, the model becomes even slower, reaching an accuracy of 5.29% and a loss of 17.00, while being able to recognize 5 stations.

Scenario 1:

Hyperparameters

Epochs =30

Batch size =16

Hidden layers =64

Activation function: tanh

```

epochs = 30
batch_size = 16
n_hidden = 64

timesteps = len(X_train[0])
input_dim = len(X_train[0][0])
n_classes = len(y_train[0])

model = Sequential()
model.add(Conv1D(n_hidden, kernel_size=2, activation='relu', input_shape=(timesteps, input_dim)))
model.add(MaxPooling1D())
model.add(LSTM(n_hidden, input_shape=(timesteps, input_dim)))
model.add(Dropout(0.5))
model.add(Dense(n_classes, activation='tanh')) # Don't use relu here!

```

Output:

```

Epoch 1/30
1076/1076 - 2s - 2ms/step - accuracy: 0.0866 - loss: 25.2647
Epoch 2/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1907 - loss: 24.6085
Epoch 3/30
1076/1076 - 2s - 1ms/step - accuracy: 0.1543 - loss: 24.9159
Epoch 4/30
1076/1076 - 2s - 2ms/step - accuracy: 0.2123 - loss: 24.8038
Epoch 5/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1932 - loss: 24.5199
Epoch 6/30
1076/1076 - 2s - 2ms/step - accuracy: 0.2373 - loss: 24.9833
Epoch 7/30
1076/1076 - 2s - 1ms/step - accuracy: 0.1938 - loss: 24.8887
Epoch 8/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1577 - loss: 24.5110
Epoch 9/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1063 - loss: 24.0851
Epoch 10/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1667 - loss: 24.8757
Epoch 11/30
1076/1076 - 2s - 2ms/step - accuracy: 0.2267 - loss: 24.8200
Epoch 12/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1931 - loss: 24.7648
Epoch 13/30
1076/1076 - 2s - 2ms/step - accuracy: 0.0853 - loss: 24.5965
Epoch 14/30
1076/1076 - 2s - 2ms/step - accuracy: 0.0859 - loss: 24.6746
Epoch 15/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1638 - loss: 24.8987
Epoch 16/30
1076/1076 - 2s - 1ms/step - accuracy: 0.1322 - loss: 24.4474
Epoch 17/30
1076/1076 - 2s - 2ms/step - accuracy: 0.0974 - loss: 24.6236
Epoch 18/30
1076/1076 - 2s - 2ms/step - accuracy: 0.2107 - loss: 24.8791
Epoch 19/30
1076/1076 - 2s - 2ms/step - accuracy: 0.2075 - loss: 24.6161
Epoch 20/30
1076/1076 - 2s - 2ms/step - accuracy: 0.2198 - loss: 24.5026
Epoch 21/30
1076/1076 - 2s - 2ms/step - accuracy: 0.0589 - loss: 24.5047
Epoch 22/30
1076/1076 - 2s - 1ms/step - accuracy: 0.0435 - loss: 24.2501
Epoch 23/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1045 - loss: 24.3699
Epoch 24/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1084 - loss: 24.2773
Epoch 25/30
1076/1076 - 2s - 2ms/step - accuracy: 0.0790 - loss: 24.6361
Epoch 26/30
1076/1076 - 2s - 2ms/step - accuracy: 0.0469 - loss: 24.2582
Epoch 27/30
1076/1076 - 2s - 2ms/step - accuracy: 0.0800 - loss: 24.5896
Epoch 28/30

```

1076/1076 - 2s - 2ms/step - accuracy: 0.0788 - loss: 24.5523
Epoch 29/30
1076/1076 - 2s - 2ms/step - accuracy: 0.0543 - loss: 24.5272
Epoch 30/30
1076/1076 - 2s - 2ms/step - accuracy: 0.0410 - loss: 24.9091

Confusion Matrix:

180/180 ————— **0s 961us/step**

Pred	DEBILT
True	
BASEL	3682
BELGRADE	1092
BUDAPEST	214
DEBILT	82
DUSSELDORF	29
HEATHROW	82
KASSEL	11
LJUBLJANA	61
MAASTRICHT	9
MADRID	458
MUNCHENB	8
OSLO	5
STOCKHOLM	4
VALENTIA	1

Scenario 2:

Hyperparameters

Epochs =30

Batch size =16

Hidden layers =4

Activation function: Softmax

```
epochs = 30
batch_size = 16
n_hidden = 4

timesteps = len(X_train[0])
input_dim = len(X_train[0][0])
n_classes = len(y_train[0])

model = Sequential()
model.add(Conv1D(n_hidden, kernel_size=2, activation='relu', input_shape=(timesteps, input_dim)))
model.add(Dense(16, activation='relu'))
model.add(MaxPooling1D())
model.add(Flatten())
model.add(Dense(n_classes, activation='softmax')) # Options: sigmoid, tanh, softmax, relu
```

Output:

Epoch 1/30
1076/1076 - 3s - 3ms/step - accuracy: 0.1138 - loss: 1012.3018
Epoch 2/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1125 - loss: 10250.8691
Epoch 3/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1121 - loss: 35854.5156
Epoch 4/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1138 - loss: 80724.9531
Epoch 5/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1094 - loss: 141010.6250
Epoch 6/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1101 - loss: 210911.2812
Epoch 7/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1131 - loss: 302564.5625
Epoch 8/30

1076/1076 - 2s - 2ms/step - accuracy: 0.1114 - loss: 404418.8125
Epoch 9/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1058 - loss: 522753.4375
Epoch 10/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1107 - loss: 655927.0000
Epoch 11/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1077 - loss: 819333.3125
Epoch 12/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1081 - loss: 1000039.1875
Epoch 13/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1087 - loss: 1201817.0000
Epoch 14/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1073 - loss: 1417414.8750
Epoch 15/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1125 - loss: 1686958.2500
Epoch 16/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1129 - loss: 1940106.3750
Epoch 17/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1101 - loss: 2232940.7500
Epoch 18/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1074 - loss: 2570307.2500
Epoch 19/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1073 - loss: 2922072.0000
Epoch 20/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1102 - loss: 3309082.5000
Epoch 21/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1082 - loss: 3714033.5000
Epoch 22/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1094 - loss: 4149961.5000
Epoch 23/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1108 - loss: 4600298.5000
Epoch 24/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1093 - loss: 5109992.0000
Epoch 25/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1134 - loss: 5635078.0000
Epoch 26/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1106 - loss: 6191976.0000
Epoch 27/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1102 - loss: 6781577.5000
Epoch 28/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1113 - loss: 7497301.0000
Epoch 29/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1088 - loss: 8194229.5000
Epoch 30/30
1076/1076 - 2s - 2ms/step - accuracy: 0.1107 - loss: 8954767.0000

Confusion Matrix:

180/180	0s 1ms/step						
Pred	BASEL	BUDAPEST	KASSEL	LJUBLJANA	MADRID	MUNCHENB	STOCKHOLM
True							
BASEL	3063	104	440	1	42	24	8
BELGRADE	999	30	63	0	0	0	0
BUDAPEST	201	8	5	0	0	0	0
DEBILT	81	0	1	0	0	0	0
DUSSELDORF	29	0	0	0	0	0	0
HEATHROW	81	0	1	0	0	0	0
KASSEL	11	0	0	0	0	0	0
LJUBLJANA	59	1	1	0	0	0	0
MAASTRICHT	9	0	0	0	0	0	0
MADRID	455	2	0	0	1	0	0
MUNCHENB	8	0	0	0	0	0	0
OSLO	4	0	1	0	0	0	0
STOCKHOLM	4	0	0	0	0	0	0
VALENTIA	1	0	0	0	0	0	0

Scenario 3:

Hyperparameters:

Epochs = 30

Batch size = 16

Hidden Layers = 64

Activation function: sigmoid

```
epochs = 30
batch_size = 16
n_hidden = 64

timesteps = len(X_train[0])
input_dim = len(X_train[0][0])
n_classes = len(y_train[0])

model = Sequential()
model.add(Conv1D(n_hidden, kernel_size=2, activation='relu', input_shape=(timesteps, input_dim)))
model.add(Dense(16, activation='relu'))
model.add(MaxPooling1D())
model.add(Flatten())
model.add(Dense(n_classes, activation='sigmoid')) # Options: sigmoid, tanh, softmax, relu
```

Output:

Epoch 1/30
1076/1076 - 1s - 845us/step - accuracy: 0.6221 - loss: 9617.2656
Epoch 2/30
1076/1076 - 1s - 552us/step - accuracy: 0.6432 - loss: 99427.4766
Epoch 3/30
1076/1076 - 1s - 566us/step - accuracy: 0.6434 - loss: 319140.3438
Epoch 4/30
1076/1076 - 1s - 572us/step - accuracy: 0.6434 - loss: 725537.8750
Epoch 5/30
1076/1076 - 1s - 577us/step - accuracy: 0.6434 - loss: 1304382.8750
Epoch 6/30
1076/1076 - 1s - 560us/step - accuracy: 0.6434 - loss: 2091254.7500
Epoch 7/30
1076/1076 - 1s - 554us/step - accuracy: 0.6435 - loss: 3066805.5000
Epoch 8/30
1076/1076 - 1s - 555us/step - accuracy: 0.6436 - loss: 4196246.5000
Epoch 9/30
1076/1076 - 1s - 595us/step - accuracy: 0.6437 - loss: 5708821.5000
Epoch 10/30
1076/1076 - 1s - 551us/step - accuracy: 0.6437 - loss: 7328497.0000
Epoch 11/30
1076/1076 - 1s - 554us/step - accuracy: 0.6437 - loss: 9340031.0000
Epoch 12/30
1076/1076 - 1s - 554us/step - accuracy: 0.6437 - loss: 11615387.0000
Epoch 13/30
1076/1076 - 1s - 554us/step - accuracy: 0.6437 - loss: 14374781.0000
Epoch 14/30
1076/1076 - 1s - 580us/step - accuracy: 0.6437 - loss: 17320762.0000
Epoch 15/30
1076/1076 - 1s - 567us/step - accuracy: 0.6437 - loss: 20846166.0000
Epoch 16/30
1076/1076 - 1s - 675us/step - accuracy: 0.6437 - loss: 24717202.0000
Epoch 17/30
1076/1076 - 1s - 583us/step - accuracy: 0.6437 - loss: 28892162.0000
Epoch 18/30
1076/1076 - 1s - 558us/step - accuracy: 0.6437 - loss: 33615856.0000
Epoch 19/30
1076/1076 - 1s - 585us/step - accuracy: 0.6437 - loss: 38488988.0000
Epoch 20/30
1076/1076 - 1s - 610us/step - accuracy: 0.6437 - loss: 44432808.0000

Epoch 21/30
1076/1076 - 1s - 560us/step - accuracy: 0.6437 - loss: 50142788.0000
Epoch 22/30
1076/1076 - 1s - 552us/step - accuracy: 0.6437 - loss: 56840280.0000
Epoch 23/30
1076/1076 - 1s - 609us/step - accuracy: 0.6437 - loss: 64049112.0000
Epoch 24/30
1076/1076 - 1s - 596us/step - accuracy: 0.6437 - loss: 71829424.0000
Epoch 25/30
1076/1076 - 1s - 628us/step - accuracy: 0.6438 - loss: 80094520.0000
Epoch 26/30
1076/1076 - 1s - 577us/step - accuracy: 0.6438 - loss: 88352840.0000
Epoch 27/30
1076/1076 - 1s - 561us/step - accuracy: 0.6438 - loss: 97845552.0000
Epoch 28/30
1076/1076 - 1s - 562us/step - accuracy: 0.6438 - loss: 108565176.0000
Epoch 29/30
1076/1076 - 1s - 560us/step - accuracy: 0.6438 - loss: 119060784.0000
Epoch 30/30
1076/1076 - 1s - 623us/step - accuracy: 0.6438 - loss: 130606680.0000

Confusion Matrix:

180/180  **0s 472us/step**

Pred	BASEL
True	
BASEL	3682
BELGRADE	1092
BUDAPEST	214
DEBILT	82
DUSSELDORF	29
HEATHROW	82
KASSEL	11
LJUBLJANA	61
MAASTRICHT	9
MADRID	458
MUNCHENB	8
OSLO	5
STOCKHOLM	4
VALENTIA	1